

Lookout 2: Anomaly Detection via Leave-One-Out Kernel Density Estimation

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September 18, 2026

Abstract

The lookout algorithm identifies anomalies by fitting a generalized Pareto distribution to the largest surprisals of a leave-one-out kernel density estimate, with a bandwidth chosen from the data by persistent homology. The death diameters it uses are the edge lengths of the Euclidean minimum spanning tree of the data, and we use the tree to analyse the algorithm and to modify it. The bandwidth becomes an upper quantile of the edge lengths in place of the largest gap between consecutive edge lengths. Using limit theory for random geometric graphs, we show that the scaled quantile converges in probability to a constant for any density, so the bandwidth has the order of the spacing between neighbouring observations in the sparse part of the sample, whereas the longest edges grow without bound under heavy tails. The resulting density estimate is not consistent, and we argue that this is appropriate for anomaly detection. We also replace min-max scaling with a robust covariance standardization, and constrain the shape parameter of the generalized Pareto distribution. In simulations and two applications the modified algorithm improves on the original in most cases and compares well with other methods.

Keywords: bandwidth selection, extreme value theory, generalized Pareto distribution, minimum spanning tree, outlier detection.

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1 Introduction

Anomaly detection is used in many application domains such as credit card fraud and scam detection (Kolupuri et al. 2025). Kandanaarachchi & Hyndman (2022) proposed the *lookout* algorithm for anomaly detection using a kernel density estimate with bandwidth based on persistent homology. Anomalies are identified by fitting a generalized Pareto distribution (GPD) to the negative logarithm of the smallest leave-one-out kernel density estimates, and declaring points with low GPD probabilities as anomalies.

In this paper we study the bandwidth of the lookout algorithm and what it implies for the kernel density estimates. The bandwidth was originally expressed in terms of the death diameters of a Vietoris-Rips filtration; these are the edge lengths of the Euclidean minimum spanning tree of the data, and we work with the tree throughout. This allows us to analyse the asymptotic behaviour of the bandwidth, and to propose a modified algorithm that takes its bandwidth from a different feature of the same edge lengths. The modified algorithm is more robust than the original and improves on its performance in most cases.

1.1 Anomalies and surprisals

There is no broadly accepted definition for an anomaly. Hawkins (1980) defines an anomaly or outlier as *an observation which deviates so much from other observations as to arouse suspicion it was generated by a different mechanism*. Barnett & Lewis (1978) define an anomaly or outlier as *an observation (or a subset of observations) which appears to be inconsistent with the remainder of that set of data*. Both these definitions agree that anomalies are rare and different from other observations. In practice, this means fitting a probability distribution to the observations and asking whether each observation is a likely draw from it. Hyndman & Frazier (2026) proposed a formalization of this concept, which we state in the following definition. Throughout, f is a probability density function on $\mathbb{R}^m$ with respect to Lebesgue measure. It is the density against which the observations are judged. We do not assume that every observation is a draw from f ;

the anomalies are the observations that are unlikely under f .

Definition 1 (Anomalies). Given a set of observations $\mathbf{y}_i \in \mathbb{R}^m$ for $i \in \{1, \dots, n\}$, and a probability density function f , we define observation $\mathbf{y}_i$ as anomalous if

$$\mathbb{P}(f(\mathbf{Y}) \leq f(\mathbf{y}_i)) < \alpha, \quad (1)$$

where $\mathbf{Y}$ is a random vector with density f , and $\alpha > 0$ is a chosen threshold.

Definition 2 (Surprisals). Given a probability density function f , the *surprisal* of an observation $\mathbf{y}$ with respect to f is defined as $-\log f(\mathbf{y})$.

These are better known as “log scores” in statistics (Good 1952), but we prefer the term “surprisals” from information theory (Stone 2022), as it provides a measure of how surprising an observation is, given the probability density function f . Specifically, it takes large values for observations that are unlikely under f .

Let $S = -\log f(\mathbf{Y})$ be the random variable representing the surprisal of a random vector $\mathbf{Y}$ with density f , and let $G(s) = \mathbb{P}(S \leq s)$ be its distribution function. Then the surprisal probability of observation $\mathbf{y}_i$ is

$$p_i = \mathbb{P}(S \geq s_i) = 1 - G(s_i^-),$$

where $s_i = -\log f(\mathbf{y}_i)$, and $G(s_i^-) = \lim_{t \uparrow s_i} G(t)$ is the left limit of G at s_i , which differs from $G(s_i)$ only when G has a jump at s_i , as happens when f is constant on a region of positive probability. An observation is an anomaly if and only if $p_i < \alpha$.

1.2 Extreme value theory

The lookout algorithm determines which surprisals are unusually large by using a peaks-over-threshold approach (Pickands 1975, Coles 2001), and fitting a GPD to the upper tail of the observed surprisals. That is,

$$\mathbb{P}(S \leq s \mid S > u) = \frac{G(s) - G(u)}{1 - G(u)} \approx 1 - (1 + \xi(s - u)/\sigma_u)^{-1/\xi}, \quad s > u, \quad 1 + \xi(s - u)/\sigma_u > 0, \quad (2)$$

where u is a threshold chosen to be large enough that the conditional distribution of the excess $S - u$ over the threshold is well approximated by a GPD, $\sigma_u > 0$ is a scale parameter depending on the threshold, and $\xi \in \mathbb{R}$ is a shape parameter. The case $\xi = 0$ is read as the limiting case $1 - \exp(-(s - u)/\sigma_u)$.

The distribution in (2) is unbounded above when $\xi \geq 0$, and has finite upper endpoint $u + \sigma_u/|\xi|$ when $\xi < 0$, so a quantity that is bounded above cannot have a positive shape parameter. We use this in Section 3 to constrain the fitted GPD.

1.3 Kernel density estimation

Given a set of observations $\mathbf{y}_i \in \mathbb{R}^m$ for $i \in \{1, \dots, n\}$, a kernel density estimate is defined as (Chacón & Duong 2018)

$$\hat{f}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n |\mathbf{H}|^{-1/2} K(\mathbf{H}^{-1/2}(\mathbf{y} - \mathbf{y}_i)), \quad (3)$$

where K is a kernel function and $\mathbf{H}$ is a symmetric $m \times m$ positive-definite bandwidth matrix. When the observations are n independent draws from f , the accuracy of $\hat{f}$ is conventionally measured by the mean integrated squared error (MISE),

$$\text{MISE}(\hat{f}) = \mathbb{E} \left[\int (\hat{f}(\mathbf{u}) - f(\mathbf{u}))^2 d\mathbf{u} \right], \quad (4)$$

where the expectation is taken over that sample, and $\hat{f}$ is a *consistent estimator* of f if $\text{MISE}(\hat{f}) \rightarrow 0$ as $n \rightarrow \infty$. We consider matrices of the form $\mathbf{H} = ah_n^2 \mathbf{I}_m$ for a fixed constant $a > 0$, where h_n is a length scale computed from the sample, so that $|\mathbf{H}|^{1/2} = a^{m/2} h_n^m$.

Definition 3 (Admissible bandwidth). We say that h_n is an *admissible bandwidth* for samples in $\mathbb{R}^m$ if $h_n \rightarrow 0$ and $nh_n^m \rightarrow \infty$ as $n \rightarrow \infty$ in probability.

For an admissible bandwidth, $\hat{f}$ is a consistent estimator of f whenever f is square integrable and twice differentiable with bounded, continuous and square integrable second derivatives, and K is a square integrable, spherically symmetric density with a finite second moment (Chacón & Duong 2018, pp. 31–37).

1.4 The lookout algorithm for anomaly detection

The *lookout* algorithm of [Kandanaarachchi & Hyndman \(2022\)](#), as implemented in the *lookout* package, has four inputs: the dataset $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, where each $\mathbf{y}_i \in \mathbb{R}^m$; a parameter $\beta \in (0, 1)$ such that the GPD is fitted to the largest $1 - \beta$ of the surprisals, with $\beta = 0.9$ following [Bommier \(2014\)](#); a parameter $\alpha \in (0, 1)$ signifying the cut-off threshold for determining an anomaly; and a logical variable *unitize* indicating whether to scale the data. We briefly outline the steps below.

Algorithm 1 Lookout algorithm

Require: Data $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, proportion $\beta \in (0, 1)$, threshold $\alpha \in (0, 1)$, *unitize* $\in \{\text{True}, \text{False}\}$.

- 1: If *unitize*, scale the data to lie in the unit cube in $\mathbb{R}^m$.
 - 2: Compute the persistent homology barcode of the (possibly scaled) data cloud for dimension zero using the Vietoris-Rips diameter ([Ghrist 2008](#)).
 - 3: From the barcode, obtain the ordered finite death diameters $\{d_i\}_{i=1}^{n-1}$ for connected components, and their successive differences $\Delta d_i = d_{i+1} - d_i$.
 - 4: Let $d_* = d_{i_*}$ where $i_* = \arg \max \{\Delta d_i\}_{i=1}^{n-2}$; that is, d_* is the diameter corresponding to the lower point of the maximum Δd_i .
 - 5: Compute the kernel density estimates at the observations: $f_i = \hat{f}(\mathbf{y}_i)$, $i = 1, \dots, n$, where $\hat{f}$ is given by (3) with the kernel $K(\mathbf{u}) = \frac{m+2}{2 \cdot 5^{m/2} b_m} (1 - \|\mathbf{u}\|^2/5)_+$, b_m being the volume of the unit ball in $\mathbb{R}^m$, and $\mathbf{H} = d_*^2 \mathbf{I}_m$, so that the kernel has support radius $\sqrt{5} d_*$.
 - 6: Compute the leave-one-out estimates $f_{-i} = \frac{1}{n-1} (n f_i - |\mathbf{H}|^{-1/2} K(\mathbf{0}))$ for $i = 1, \dots, n$.
 - 7: Fit a GPD to the largest $1 - \beta$ of the surprisals $\{-\log f_i\}_{i=1}^n$, obtaining estimates of the location, scale and shape parameters (μ, σ, ξ) .
 - 8: Calculate the probability of each observation by applying the estimated GPD to the leave-one-out estimates: $p_i = (1 - \beta)(1 - \hat{F}(-\log f_{-i}))$, where $\hat{F}$ is the distribution function of the fitted GPD.
 - 9: If $p_i < \alpha$, declare $\mathbf{y}_i$ anomalous.
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Two details of [Algorithm 1](#) differ from the published description. From version 0.1.4 of the

lookout package, the search in step 4 is restricted to the death diameters at or above their median, so that a large gap among the shortest diameters is not chosen. Equation (5) of [Kandanaarachchi & Hyndman \(2022\)](#) gives the bandwidth matrix as $d_*^{2/m} \mathbf{I}_m$, whereas the implementation uses $d_*^2 \mathbf{I}_m$ in every dimension, as in step 5.

The lookout algorithm provides density-based anomaly detection, where anomalies are identified as points with low density in the space of the data. If there are no true anomalies in the data, the algorithm should identify a proportion α of the data as false positive “anomalies”.

Steps 2–4 of [Algorithm 1](#) can be expressed in terms of the Euclidean minimum spanning tree T of the data. Components of the Vietoris-Rips complex merge when the shortest edge between them appears, which is Kruskal’s algorithm, so the merges occur along the edges of T . The finite death diameters $d_1 \leq \dots \leq d_{n-1}$ are therefore the ordered edge lengths $e_{(1)} \leq \dots \leq e_{(n-1)}$ of T ([Adams et al. 2020](#), Remark 1), and also the merge heights of the single linkage dendrogram ([Gower & Ross 1969](#)). In the rest of the paper we use the tree rather than the filtration, so that we can apply the limit theory for minimum spanning trees.

While [Kandanaarachchi & Hyndman \(2022\)](#) demonstrated excellent results in applying their algorithm to a variety of simulated and real datasets, we see several potential areas for improvement that we aim to address in this paper. First, elements of the algorithm are not robust to outliers, including the min-max scaling, and the use of the largest gap in the Rips death diameters (equivalently, edge lengths). Second, the diagonal bandwidth matrix $\mathbf{H}$ does not account for correlations between variables, which can lead to suboptimal performance if the data have strong correlations. Third, it is not obvious that d_* is an admissible bandwidth, or even that it shrinks to zero, particularly for distributions with heavy tails. Finally, the GPD estimation made minimal assumptions about the distribution of the surprisal values, but we know that these are bounded above and below, and that fact can be used to improve the estimated GPD.

1.5 The modified lookout algorithm

As a result of these considerations, we propose three modifications to the lookout algorithm, described in detail in [Section 3](#).

1. Rather than use a min-max scaling of the data, we standardize the data using a robust covariance estimator. That is, we scale and rotate the data to ensure that the covariance matrix of the transformed data is approximately the identity matrix. This ensures that the data have zero pairwise correlations, which leads to more efficient kernel density estimates. It also leads to each variable having approximately unit variance, which provides a more robust scaling than min-max scaling.
2. Rather than using the lower point of the largest gap in the minimum spanning tree edge lengths, we use an upper quantile of those edge lengths. This avoids the problem that a few anomalies at the top of the edge length distribution can determine the largest gap and hence the bandwidth. It also gives the bandwidth the order $n^{-1/m}$ of the spacing between neighbouring observations, for any density.
3. The estimated GPD shape parameter is constrained to be non-positive, which improves the stability of the GPD fitting. This is justified by noting that the kernel density estimates are bounded away from zero, so the surprisals are bounded above and the fitted GPD cannot have a positive shape parameter.

Note that we do not need to estimate the density of the underlying distribution directly. Instead, we can estimate the density of a transformation of the data, and identify anomalies in the transformed space. Hence, there is no need to undo the scaling or rotation.

1.6 Outline of the paper

The remainder of this paper is organized as follows. In [Section 2](#), we define the bandwidth as a quantile of the edge lengths of the Euclidean minimum spanning tree, and relate it to existing

minimum spanning tree and nearest-neighbour methods. We also establish the order of that bandwidth for any density, show that the longest edges, among which the original bandwidth typically lies, can grow without bound, and examine what the order implies for the kernel density estimates. In [Section 3](#), we propose a modified lookout algorithm that improves performance and robustness under a broader range of conditions. In [Section 4](#), we report five simulation experiments, comparing the modified algorithm first with the original lookout and then with four other anomaly detection methods, and examine the sensitivity of the results to the bandwidth. [Section 5](#) applies both versions to two real datasets. Finally, in [Section 6](#), we summarize our results and discuss future work.

2 Bandwidth selection from the minimum spanning tree

2.1 The bandwidth rule

Let T denote the Euclidean minimum spanning tree of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\} \subset \mathbb{R}^m$: the connected graph on the n observations whose total edge length is smallest. Provided the pairwise distances are distinct, which holds almost surely when the observations are drawn from a continuous density, T is unique and has $n - 1$ edges. When placed in increasing order, the edge lengths are denoted by $e_{(1)} \leq \dots \leq e_{(n-1)}$.

Definition 4 (Minimum spanning tree bandwidth). For $\gamma \in (0, 1)$, the *minimum spanning tree bandwidth* is the γ quantile of the edge lengths of T ,

$$h_n = e_{(\lceil \gamma(n-1) \rceil)}. \quad (5)$$

We take $\gamma = 0.98$ throughout unless stated otherwise. In computation we use the median-unbiased quantile estimator of [Hyndman & Fan \(1996\)](#) rather than the order statistic itself; the difference is negligible for large n . The kernel density estimate [\(3\)](#) is then computed with the

bandwidth matrix

$$\mathbf{H} = ah_n^2 \mathbf{I}_m, \quad (6)$$

where $a = m + 4$ for the Epanechnikov kernel, which makes h_n the per-coordinate standard deviation of the kernel.

Equation (5) has an interpretation in terms of single linkage clustering. Cutting the single linkage dendrogram at height h_n leaves about $(1 - \gamma)n$ clusters. With $\gamma = 0.98$, h_n is the height at which 2% of the merges remain to be made, so at most 2% of the observations are still isolated. It is the largest distance between neighbouring observations once the sparsest 2% of the sample is excluded.

To compute h_n , we use the dual-tree Borůvka algorithm of [March et al. \(2010\)](#), as implemented in `mlpack` ([Singh Parihar et al. 2026](#)), which builds the exact Euclidean minimum spanning tree in $O(n \log n)$ time under the same assumptions that make dual-tree nearest-neighbour search efficient.

[Fig. 1](#) shows the rule on a sample of 300 standard normal observations in $\mathbb{R}^2$ with three anomalies added. The left panel shows the minimum spanning tree, with the edges longer than h_n drawn in colour; the middle panel shows the single linkage dendrogram with the cut at h_n ; the right panel shows the kernel density estimate computed with the bandwidth matrix (6) and the observations declared anomalous by [Algorithm 2](#), which include two of the 300 normal observations, as expected with $\alpha = 0.01$.

[Table 1](#) gives the time taken by the tree and by the whole of [Algorithm 2](#) for standard normal samples, on a desktop computer with an Intel Core Ultra 9 285 processor. The tree takes under four seconds in every case except $m = 10$ at $n = 10^5$, where it takes a minute. The kernel density estimate is computed either exactly, by a fixed-radius search whose cost is proportional to the number of pairs of observations inside the kernel support, or by a truncated sum over at most 500 nearest neighbours of each observation. In two dimensions the support holds few observations and the exact estimate takes under two seconds at $n = 10^5$. In five and ten dimensions the support in the bulk of the data holds a large fraction of the sample ([Table 2](#)), so the exact estimate costs of

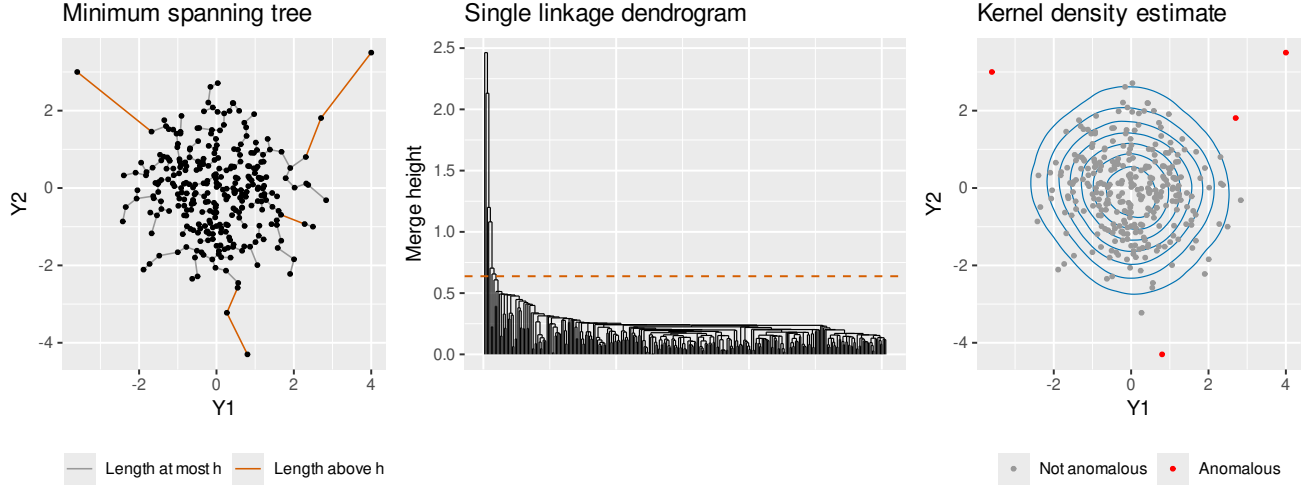


Figure 1: The minimum spanning tree bandwidth for 300 standard normal observations in $\mathbb{R}^2$ and three anomalies. Left: the minimum spanning tree; edges longer than h_n are shown in colour. Middle: the single linkage dendrogram, with the dashed line at height h_n . Right: contours of the kernel density estimate with bandwidth matrix (6), and the observations identified as anomalous in red.

order n^2 and is run only up to $n = 10^4$; the truncated sum then takes 18 seconds for $n = 10^5$ in five dimensions and two and a half minutes in ten, and 99 seconds for $n = 10^6$ in two. In Experiments 3 to 5 of Section 4, where $n \leq 805$, a run of lookout takes about 0.1 seconds or less; KDEOS takes about the same, stray and HDoutliers less, and RDOS one to five seconds.

2.2 Other uses of the minimum spanning tree

Minimum spanning trees have been used in multivariate statistics for many years, including for anomaly detection. Rohlf (1975) gave the classical minimum spanning tree test for multivariate outliers. When $m = 1$, the edges of the minimum spanning tree are the gaps between consecutive order statistics, so the edge lengths of the tree generalize the gaps of the univariate gap test; Rohlf tests the largest squared edge length, divided by the mean squared edge length, with critical values derived from a gamma distribution for the squared edge lengths. However, Caroni & Prescott

Table 1: Seconds taken by the minimum spanning tree and by the whole of [Algorithm 2](#) for standard normal samples in $\mathbb{R}^m$, on a desktop computer with an Intel Core Ultra 9 285 processor. The exact leave-one-out estimate uses a fixed-radius neighbour search and was run for $n \leq 10^4$, and up to $n = 10^5$ in two dimensions; the truncated estimate sums each kernel over at most 500 nearest neighbours.

m	n	Tree	Total, exact KDE	Total, truncated KDE
2	10^3	0.012	0.045	0.062
5	10^3	0.009	0.074	0.076
10	10^3	0.043	0.14	0.14
2	10^4	0.023	0.28	0.16
5	10^4	0.16	2.9	0.47
10	10^4	2	5.1	3.2
2	10^5	0.29	1.4	5.5
5	10^5	2.3		18
10	10^5	60		147
2	10^6	3.3		99

([1995](#)) argue that this is not a useful formal test.

Related uses of the tree include single linkage detectors, which exploit the equivalence between the minimum spanning tree and single linkage clustering ([Gower & Ross 1969](#)) to cut the dendrogram and declare the resulting small clusters anomalous ([Jiang et al. 2001](#)), and minimum spanning tree clustering, which deletes inconsistently long edges to recover cluster structure ([Zahn 1971](#)). [Friedman & Rafsky \(1979\)](#) give the broader statistical setting, using minimum spanning trees to construct multivariate analogues of the Wald-Wolfowitz and Smirnov two-sample tests. [Juszczak et al. \(2009\)](#) build a one-class classifier from the tree of a training sample, and [Ahmed](#)

et al. (2019) replace Euclidean distances by distances along the tree in a local outlier score.

The tree has also been used to choose the bandwidth of a kernel density estimate. Chaudhuri et al. (1996) obtain the bandwidth from the tree of the sample and show that the resulting estimate with a uniform kernel is asymptotically unbiased on bounded sets. Sreevani & Murthy (2016) define the bandwidth as a function of the total length of the tree, with outlier rejection among the applications.

Our use of the tree differs from all of these. We use it only to choose the bandwidth of the kernel density estimate, and anomalies are identified by the leave-one-out kernel density estimate and the fitted generalized Pareto distribution, as in Algorithm 1, so a long edge of T does not by itself make its endpoint an anomaly, and an observation can be anomalous without being incident to a long edge. Unlike the total length of the tree, which is determined by the bulk of the data, the quantile in Definition 4 is determined by the sparsest part of the sample.

2.3 The order of the quantile edge length

We study the quantile through its relationship to single linkage clustering. For $r > 0$, let $C_n(r)$ be the number of connected components of the random geometric graph that joins every pair of observations at distance at most r . Kruskal’s algorithm (Kruskal 1956) builds T by adding edges in increasing order of length, so the edges of T of length at most r form a spanning forest of that graph, with one tree for each component (Gower & Ross 1969). There are therefore $n - C_n(r)$ such edges, and $h_n \leq r$ exactly when $C_n(r) \leq n - \lceil \gamma(n - 1) \rceil$.

Component counts of random geometric graphs obey a law of large numbers (Penrose 2003, Theorem 13.25), which Mitsche & Penrose (2021) extend to points drawn from any distribution. The limit involves a Poisson process. For $\lambda > 0$, let $\mathcal{P}_\lambda$ be a homogeneous Poisson process of intensity λ on $\mathbb{R}^m$, join each pair of points of $\mathcal{P}_\lambda \cup \{\mathbf{0}\}$ at distance at most 1, let $\mathcal{C}_\lambda$ be the component containing the origin, and put $\kappa(\lambda) = \mathbb{E}[1/|\mathcal{C}_\lambda|]$, with $1/|\mathcal{C}_\lambda| = 0$ when the component is infinite. Thus $\kappa(\lambda)$ is the expected number of components per point in a large sample when

points of intensity λ are joined at unit distance.

Proposition 1. *Let $\mathbf{y}_1, \dots, \mathbf{y}_n$ be sampled independently from a density f on $\mathbb{R}^m$, and fix $\gamma \in (0, 1)$.*

Then $n^{1/m}h_n \rightarrow q_\gamma(f)$ in probability as $n \rightarrow \infty$, where $q_\gamma(f) \in (0, \infty)$ is the unique solution q of

$$\int_{\mathbb{R}^m} \kappa(q^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x} = 1 - \gamma. \quad (7)$$

Moreover $t_- \leq q_\gamma(f) \leq t_+$, where t_- and t_+ are the unique positive solutions t of

$$\int_{\mathbb{R}^m} \exp(-b_m t^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x} = 1 - \gamma \quad \text{and} \quad \int_{\mathbb{R}^m} (1 - \exp(-b_m t^m f(\mathbf{x}))) d\mathbf{x} = (1 - \gamma) b_m t^m$$

respectively, and b_m is the volume of the unit ball in $\mathbb{R}^m$.

The proof is in the online Appendix. No smoothness, boundedness or tail condition on f is required. Equation (7) is the single linkage interpretation in the limit: after rescaling distances by $n^{1/m}$, $q_\gamma(f)$ is the scale at which a large sample from f has $1 - \gamma$ components per observation.

The bounds t_- and t_+ follow from the inequalities $\exp(-b_m \lambda) \leq \kappa(\lambda) \leq (1 - \exp(-b_m \lambda))/(b_m \lambda)$: the origin forms a component on its own with probability $\exp(-b_m \lambda)$, and its component always contains its neighbours, whose number is Poisson with mean $b_m \lambda$. The lower bound also relates the tree to nearest-neighbour distances, which have the same $n^{-1/m}$ scaling. By the cut property, every nearest-neighbour distance is an edge length of T . Moreover, each component of the graph at scale r with $k \geq 2$ observations contains $k - 1$ edges of T of length at most r but k observations whose nearest neighbour lies within r , and a component with one observation contains neither. Hence $h_n \geq D_{(\lceil \gamma(n-1) \rceil)}$ for every sample, where $D_{(1)} \leq \dots \leq D_{(n)}$ are the ordered nearest-neighbour distances, and t_- is the limit of $n^{1/m} D_{(\lceil \gamma(n-1) \rceil)}$. The difference between t_- and $q_\gamma(f)$ is due to the components with more than one observation: the tree records how the observations are connected to each other, whereas nearest-neighbour distances record only the distance from each observation to its nearest neighbour.

Proposition 1 does not cover the longest edge $e_{(n-1)}$, whose limiting distribution is known for uniform points in the unit square (Penrose 1997), for normally distributed points (Penrose 1998),

and for a range of distributions on $\mathbb{R}^2$ with unbounded support (Hsing & Rootzén 2005). In each case the largest nearest-neighbour distance has the same limiting distribution.

2.4 A fixed quantile versus the largest gap

The original rule of Algorithm 1 takes the lower end of the largest gap between consecutive edge lengths. Under heavy tails that gap typically lies among the few longest edges, where Proposition 1 says nothing and the edge lengths can grow without bound.

Counterexample 1. *Let the observations be sampled independently from the density $f(\mathbf{x}) = c_m/(1 + \|\mathbf{x}\|^{m+1})$ on $\mathbb{R}^m$, $m \geq 1$, and put $\omega_n = n - \lfloor n^{1/4}/4 \rfloor$, which is at most $n - 1$ once $n \geq 256$. Then $\mathbb{P}(e_{(\omega_n)} > n^{1/8}) \rightarrow 1$ as $n \rightarrow \infty$. In particular $e_{(\omega_n)}$, and every longer edge of T , tends to infinity in probability.*

The proof is in the online Appendix. With probability tending to one, more than $n^{1/4}/4$ observations lie beyond radius $R_n \asymp n^{3/4}$, and no two observations beyond radius $R_n - n^{1/8}$ are within distance $n^{1/8}$ of each other. Each observation beyond R_n is then a component on its own when observations are joined at distance $n^{1/8}$, so $C_n(n^{1/8}) > n - \omega_n$: fewer than ω_n edges of T have length at most $n^{1/8}$, and $e_{(j)} > n^{1/8}$ for every $j \geq \omega_n$. Adler et al. (2014) call this behaviour crackle. Under heavy tails, isolated observations occur at increasing distances from the origin, and each of them contributes one of the longest edges of T . These edges lie above any fixed quantile once $n^{1/4}/4 < (1 - \gamma)n$, and Proposition 1 gives h_n of order $n^{-1/m}$ for this density as for any other. Under this density the longest edges violate the first condition of Definition 3, so a rule that chooses among them, as the largest-gap rule typically does, does not give an admissible bandwidth.

The Gaussian distribution leads to a similar conclusion. For Gaussian observations in $\mathbb{R}^m$ with $m \geq 2$, the longest edge of T is of order $\log \log n / \sqrt{\log n}$ (Penrose 1998), which tends to zero much more slowly than $n^{-1/m}$. A rule that chooses among the longest edges therefore has the right order only if the gap it finds lies well below the top of the edge-length distribution.

2.5 The critical scaling

We now consider the implications of (6) for the kernel density estimate.

Proposition 2. *Let f be a density on $\mathbb{R}^m$ and let $\mathbf{H} = (m+4)h_n^2 \mathbf{I}_m$ with $h_n = e_{(\lceil \gamma(n-1) \rceil)}$. Then $n|\mathbf{H}|^{1/2} \rightarrow (m+4)^{m/2} q_\gamma(f)^m$ in probability as $n \rightarrow \infty$, and the limit is finite and positive.*

Proof. $|\mathbf{H}|^{1/2} = (m+4)^{m/2} h_n^m$, so $n|\mathbf{H}|^{1/2} = (m+4)^{m/2} (n^{1/m} h_n)^m$, and Proposition 1 gives $n^{1/m} h_n \rightarrow q_\gamma(f) \in (0, \infty)$ in probability. $\square$

The quantile bandwidth is therefore not admissible: the first condition of Definition 3 holds and the second fails, since nh_n^m converges to a positive constant. The variance term $n^{-1}|\mathbf{H}|^{-1/2}R(K)$ of the MISE does not vanish, and the estimator is not MISE-consistent. The expected number of observations inside the kernel support at a point $\mathbf{x}$ is asymptotically $nb_m|\mathbf{H}|^{1/2}f(\mathbf{x})$, which by Proposition 2 converges to $b_m(m+4)^{m/2}q_\gamma(f)^m f(\mathbf{x})$. The number of observations in the window is therefore bounded as $n \rightarrow \infty$, and is proportional to the local density. Nearest-neighbour outlier scores and the local outlier factor (Breunig et al. 2000) are also based on a bounded number of neighbours, and are also inconsistent as density estimates. They fix the number of neighbours and let the radius vary, while (6) fixes the radius and lets the number of neighbours vary with f . In Section 2.6 we argue that this is appropriate for anomaly detection.

Remark 1 (Consistency by inflation). *A consistent estimator can be obtained by inflating the bandwidth. For any $a_n \rightarrow \infty$ with $a_n = o(n^{1/m})$, however slowly a_n grows, the bandwidth $a_n h_n$ satisfies $a_n h_n \rightarrow 0$ and $n(a_n h_n)^m \asymp a_n^m \rightarrow \infty$, so it is an admissible bandwidth in the sense of Definition 3 and the resulting estimator is MISE-consistent. For example, $a_n = \log \log n$ multiplies the bandwidth by a factor between 1.9 and 2.6 for n between 10^3 and 10^6 . We do not inflate the bandwidth, for the reasons given in the next section.*

2.6 The bandwidth and the MISE

An asymptotic approximation to the MISE (4) is given by [Chacón & Duong \(2018\)](#):

$$\text{AMISE}(\hat{f}) = n^{-1}|\mathbf{H}|^{-1/2}R(K) + \frac{1}{4}\mu_2(K)^2 \int \{\text{tr}(\mathbf{H}D^2f(\mathbf{u}))\}^2 d\mathbf{u}, \quad (8)$$

where $R(K) = \int K(\mathbf{u})^2 d\mathbf{u}$, the constant $\mu_2(K)$ is defined by $\int \mathbf{u}\mathbf{u}^T K(\mathbf{u}) d\mathbf{u} = \mu_2(K)\mathbf{I}_m$, and D^2f is the Hessian of f . Writing $\mathbf{H} = h^2\mathbf{I}_m$ with h a length, this is

$$\text{AMISE} = n^{-1}h^{-m}R(K) + \frac{1}{4}\mu_2(K)^2h^4 \int \{\Delta f(\mathbf{u})\}^2 d\mathbf{u}, \quad (9)$$

minimized at $h_{\text{opt}} \asymp n^{-1/(m+4)}$, with MISE of order $n^{-4/(m+4)}$. By [Proposition 1](#), $h_n \asymp n^{-1/m}$. The ratio is

$$\frac{h_n}{h_{\text{opt}}} \asymp n^{-4/\{m(m+4)\}},$$

which tends to zero for every $m \geq 1$. Asymptotically, the minimum spanning tree bandwidth undersmooths relative to MISE in every dimension: at $h \asymp n^{-1/m}$ the bias term of (9) is $O(n^{-4/m})$ while the variance term is $\Theta(1)$. However, the exponent $4/\{m(m+4)\}$ is small and the constant $q_\gamma(f)$ is large, so the asymptotic comparison is misleading at practical sample sizes. [Table 2](#) reports $n^{1/m}h_n$ for standard normal samples with $\gamma = 0.98$, the ratio of h_n to the bandwidth h_{opt} that minimizes (9) for the Epanechnikov kernel and this density, and the expected number of observations inside the kernel support at the mode. The scaled quantile changes little with n . The ratio h_n/h_{opt} is below one only for $m = 2$ and $n \geq 4000$. For $m = 3$ it is above one at $n = 16000$, and for $m = 5$ it is about three at all three sample sizes and decreases as $n^{-4/45}$. In moderate dimensions the rule therefore smooths the bulk of the data more than the MISE-optimal bandwidth does, and for $m = 5$ the window at the mode contains most of the sample.

The MISE is an integrated criterion weighted by the density, so it is dominated by the bulk of the distribution, and the bandwidth that minimizes it is determined by the spacing of observations in the bulk. Anomalies are identified from the values of $\hat{f}$ at observations where f is small, and from the upper tail of the surprisals $-\log \hat{f}(\mathbf{y}_i)$, so the relevant spacing is that of the observations

Table 2: The minimum spanning tree bandwidth for standard normal samples in $\mathbb{R}^m$ with $\gamma = 0.98$, averaged over ten samples. h_{opt} minimizes the AMISE (9) for the Epanechnikov kernel and this density, in the same units as h_n . The last column is the expected number of observations inside the kernel support at the mode, $n\mathbb{P}(\|\mathbf{Y}\| \leq \sqrt{m+4}h_n)$.

m	n	$n^{1/m}h_n$	h_n/h_{opt}	Observations in window at mode
2	1000	10.6	1.08	290
2	4000	12.1	0.77	410
2	16000	12.3	0.50	450
3	1000	7.4	2.12	720
3	4000	8.2	1.79	1,600
3	16000	8.5	1.42	2,400
5	1000	5.8	3.51	1,000
5	4000	6.1	3.24	3,900
5	16000	6.3	2.98	13,000

in the tails. A bandwidth determined by the bulk may be either too small or too large for the tails. If it is smaller than the spacing in the tails, as h_{opt} is for $m \geq 3$ in Table 2, then any observation with no other observation inside its kernel support has a leave-one-out estimate of exactly zero, and the isolated observations cannot be ranked. If it is larger, as h_{opt} is for $m = 2$ and large n , then the estimate at an isolated observation includes contributions from distant observations, and the observation no longer appears isolated. The quantile rule sets the bandwidth from the spacing in the tails. By the single linkage interpretation of (5), h_n is the scale at which all but about 2% of the observations have joined a component, so the kernel support around any observation other than the most isolated few contains at least one other observation. In the bulk, the estimate is oversmoothed at practical sample sizes and undersmoothed in the limit. This has little effect on

anomaly detection, because the estimates in the bulk are large in either case, and those observations are not identified as anomalies.

We therefore do not require consistency or an optimal rate. We require that h_n has the same order as the spacing between neighbouring observations in the sparse part of the sample, which is $n^{-1/m}$ with a constant determined by the low-density region of f through (7). [Proposition 1](#) shows that the quantile rule has this property. The largest-gap rule typically chooses among the longest edges, and under the heavy tails of [Counterexample 1](#) these grow without bound. In [Section 4.6](#) we examine the sensitivity of the results to the bandwidth, by scaling h_n up and down and by replacing it with h_{opt} .

3 Lookout algorithm v2

In light of the preceding theoretical considerations, we propose three modifications to the lookout algorithm, described in this section.

3.1 Standardization

The *minimum covariance determinant* estimator provides a robust estimate of Σ , the covariance matrix of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, by searching for the subsample of a given size whose sample covariance matrix has the smallest determinant ([Rousseeuw & Driessen 1999](#)). We take that subsample to be of size $0.9n$. This has a lower breakdown point than the usual choice of $n/2$, but anomalies are assumed to be rare, and the larger subsample gives a more efficient estimate of Σ for the non-anomalous observations. Let

$$\hat{\Sigma}^{-1} = \mathbf{U}'\mathbf{U} \tag{10}$$

denote the Cholesky decomposition of its inverse. Then, the data can be scaled by $\mathbf{U}$, giving $\mathbf{z}_i = \mathbf{U}(\mathbf{y}_i - \mathbf{m})$, where $\mathbf{m}$ is the vector of medians of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$. The covariance matrix of the scaled data $\{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ is (approximately) the identity matrix. This can be thought of as both

scaling and rotating the data; it is the multivariate equivalent of computing robust z-scores.

We prefer this to the min-max scaling used in the original lookout algorithm because it is robust to outliers and removes the correlations between variables. The kernel in [Algorithm 2](#) also has a covariance matrix that is a multiple of the identity, so the kernel is aligned with the standardized data; alignment is not necessary, but it improves the efficiency of the kernel density estimate ([Chacón & Duong 2018](#)).

3.2 Bandwidth matrix selection

Let $e_{(1)} \leq \dots \leq e_{(n-1)}$ denote the ordered edge lengths of the Euclidean minimum spanning tree of the standardized data $\mathbf{z}_1, \dots, \mathbf{z}_n$, and let $h_n = e_{(\lceil \gamma(n-1) \rceil)}$ be the bandwidth of [Definition 4](#). The kernel density estimate is computed from the standardized data with the bandwidth matrix (6), so that by [Proposition 2](#) the number of observations inside the kernel window is bounded as $n \rightarrow \infty$, which [Section 2.6](#) argues is appropriate for anomaly detection. When the proportion of anomalies is small, we expect this estimate to be close to the density of the non-anomalous observations. An equivalent estimate for the original data would use $\mathbf{H} = (m+4)h_n^2 \hat{\Sigma}$; we work with the standardized data throughout.

3.3 Bounds for kernel density estimates

The surprisals to which the GPD is fitted are bounded, which constrains the shape parameter.

Remark 2. Let K be a kernel with $0 \leq K(\mathbf{u}) \leq K(\mathbf{0}) = K_0$ for all $\mathbf{u}$, and let $\hat{f}$ be the estimate (3) with bandwidth matrix $\mathbf{H}$. The term $i = j$ of the sum contributes $|\mathbf{H}|^{-1/2} K_0/n$ and no term exceeds that, so

$$\frac{K_0 |\mathbf{H}|^{-1/2}}{n} \leq \hat{f}(\mathbf{y}_j) \leq K_0 |\mathbf{H}|^{-1/2}, \quad j = 1, \dots, n,$$

and the surprisals $-\log \hat{f}(\mathbf{y}_j)$ lie in the interval $[\frac{1}{2} \log |\mathbf{H}| - \log K_0, \frac{1}{2} \log |\mathbf{H}| + \log(n/K_0)]$, whose endpoints depend on the sample only through $\mathbf{H}$.

For the Epanechnikov kernel $K(\mathbf{u}) = \frac{m+2}{2b_m}(1 - \|\mathbf{u}\|^2)_+$, where $u_+ = \max(0, u)$, $K_0 = (m + 2)/(2b_m)$.

A GPD with non-negative shape parameter has unbounded support, so when we fit the GPD to the largest $1 - \beta$ of $\{-\log \hat{f}(\mathbf{y}_j)\}_{j=1}^n$ we constrain its shape parameter to be non-positive, which improves the stability of the fit.

The GPD is fitted under two approximations. The surprisals are computed from $\hat{f}$ rather than from f , and they are evaluated at the observations rather than at independent draws, so the values to which the GPD is fitted are neither mutually independent nor distributed exactly as S in [Section 1](#).

3.4 Modified lookout algorithm

[Algorithm 2](#) summarises the modified algorithm.

As with the original lookout algorithm, the values of p_i can also act as anomaly scores, with lower values indicating more anomalous points. The probability p_i in step 7 is the unconditional probability of a surprisal at least as large as $-\log f_{-i}$, which is $1 - \beta$ times the exceedance probability of the fitted GPD. In step 6 the implementation fits the GPD without constraint and, if the estimated shape parameter is positive, refits it with the shape parameter fixed at zero.

The default values are $\alpha = 0.01$, $\beta = 0.90$ and $\gamma = 0.98$, with scaling applied. They are used in all experiments reported here unless stated otherwise.

Step 4 evaluates (3) at each of the n observations. The implementation finds the observations inside the kernel support of each observation by a fixed-radius search, so the cost is proportional to the number of such pairs. [Table 2](#) shows that in three or more dimensions the support around an observation in the bulk of the data can hold a large fraction of the sample, in which case that cost is of order n^2 , and the exact sum is practical only for n up to about 10^4 ([Table 1](#)). For larger n the implementation instead sums each kernel over the nearest neighbours of the observation only, between 100 and 500 of them depending on n , which underestimates $\hat{f}$ in the bulk. The

Algorithm 2 Modified lookout algorithm

Require: Data $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, proportion $\beta \in (0, 1)$, threshold $\alpha \in (0, 1)$, and quantile level $\gamma \in (0, 1)$.

- 1: Optional scaling: calculate the minimum covariance determinant estimate $\hat{\Sigma}$ and the Cholesky decomposition (10), and centre, rotate and scale the data using $\mathbf{U}$, giving $\mathbf{z}_i = \mathbf{U}(\mathbf{y}_i - \mathbf{m})$ for $i = 1, \dots, n$, where $\mathbf{m}$ is the vector of medians of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$.
 - 2: Compute the Euclidean minimum spanning tree of $\{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ (set $\mathbf{z}_i = \mathbf{y}_i$ if no scaling is applied), and obtain its ordered edge lengths $e_{(1)} \leq \dots \leq e_{(n-1)}$.
 - 3: Let $h_n = e_{(\lceil \gamma(n-1) \rceil)}$, the γ sample quantile computed from $\{e_{(i)}\}_{i=1}^{n-1}$.
 - 4: Compute the kernel density estimates at the scaled observations: $f_i = \hat{f}(\mathbf{z}_i)$, $i = 1, \dots, n$, where $\hat{f}$ is given by (3) applied to $\mathbf{z}_1, \dots, \mathbf{z}_n$ with the Epanechnikov kernel $K(\mathbf{u}) = \frac{m+2}{2b_m}(1 - \|\mathbf{u}\|^2)_+$ and $\mathbf{H} = (m+4)h_n^2 \mathbf{I}_m$.
 - 5: Compute the leave-one-out estimates $f_{-i} = \frac{1}{n-1}(nf_i - |\mathbf{H}|^{-1/2}K(\mathbf{0}))$ for $i = 1, \dots, n$.
 - 6: Fit a GPD to the largest $1 - \beta$ of the surprisals $\{-\log f_i\}_{i=1}^n$, constraining the shape parameter to be non-positive, obtaining estimates of the location, scale and shape parameters (μ, σ, ξ) .
 - 7: Calculate the probability of each observation by applying the estimated GPD to the leave-one-out estimates: $p_i = (1 - \beta)(1 - \hat{F}(-\log f_{-i}))$, where $\hat{F}$ is the distribution function of the fitted GPD.
 - 8: If $p_i < \alpha$, declare $\mathbf{y}_i$ anomalous.
-

observations relevant to anomaly detection have few neighbours inside the support, and their estimates are unchanged. The truncated sum is used in Experiment 2, in Experiment S2 of the online Appendix, and for the wine reviews data in [Section 5](#).

4 Numerical experiments

We now evaluate the new version of lookout in five simulation experiments. Experiments 1 and 2 compare it against the original version, Experiments 3 to 5 compare both versions with four other anomaly detection methods, and [Section 4.6](#) examines the sensitivity of the results to the bandwidth. Two further comparisons of the two versions, and a set of showcase examples, are in the online Appendix. Both versions are run in the current implementation, version 2.1.0 of the `lookout` package ([Kandanaarachchi et al. 2026](#)). The original version uses min-max scaling, the largest-gap bandwidth of [Algorithm 1](#) with the median restriction, and an unconstrained GPD fit, and is otherwise identical to [Algorithm 2](#): in particular, it uses $\mathbf{H} = (m + 4)d_*^2 \mathbf{I}_m$ with the kernel of [Algorithm 2](#), so that d_* plays the role of h_n , and the same α , β and probability p_i . The differences between the two versions are therefore due only to the three modifications of [Section 3](#).

The four other methods are HDoutliers ([Wilkinson 2017](#)) and *stray* ([Talagala et al. 2021](#)), which apply extreme value theory to nearest-neighbour distances and are run with their α parameter set to 0.01, the default for *stray*, and KDEOS ([Schubert et al. 2014](#)) and RDOS ([Tang & He 2017](#)), which are based on kernel density estimates and are run with their default settings. The same four methods were used for comparison by [Kandanaarachchi & Hyndman \(2022\)](#). Experiments 1 and 2 report the true positive and false negative rates of the two versions of lookout. HDoutliers and *stray* return labels, so in Experiments 3 to 5 we compare them with lookout by the Fmeasure and Gmean,

$$\text{Fmeasure} = \frac{2 \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad \text{Gmean} = \sqrt{\text{Sensitivity} \times \text{Specificity}},$$

where $\text{Precision} = \text{TP}/(\text{TP} + \text{FP})$, $\text{Recall} = \text{Sensitivity} = \text{TP}/(\text{TP} + \text{FN})$ and $\text{Specificity} =$

$TN/(TN + FP)$, with TP, FP, TN and FN the numbers of true positives, false positives, true negatives and false negatives. KDEOS and RDOS return anomaly scores, so we compare them with the scores p_i of lookout by the area under the ROC curve (AUC), which plots sensitivity against one minus specificity over all thresholds.

4.1 Experiment 1: Different anomaly rates for Gamma distribution

The non-anomalous observations have independent $\text{Gamma}(2, 2)$ coordinates in $\mathbb{R}^2$, and the anomalies independent $\text{Gamma}(2, r)$ coordinates with $r \in \{0.1, 0.2, \dots, 1\}$. Each replication has 500 non-anomalies and 10 anomalies, and there are 10 replications for each value of r .

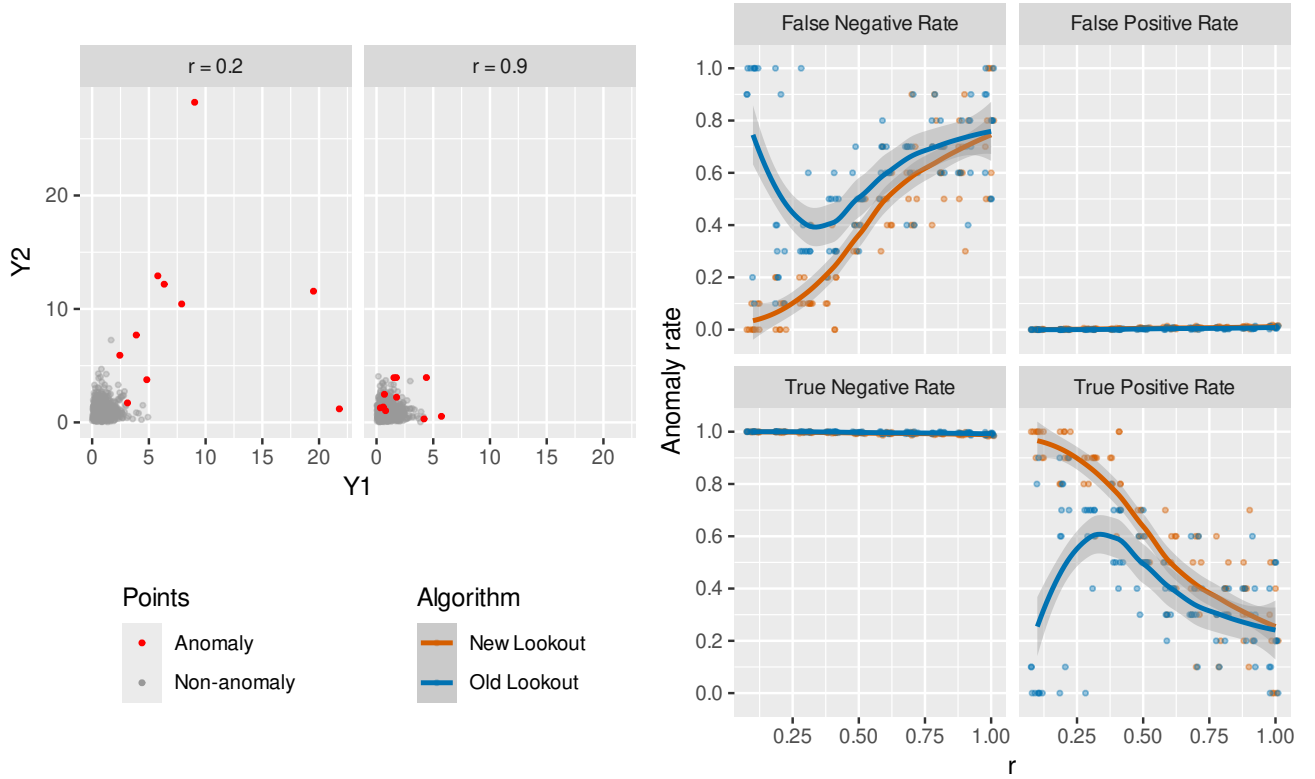


Figure 2: Experiment 1 data and results. Left: data for two iterations of the experiment with $r = 0.2$ and $r = 0.9$. Anomalies with lower values of r are easier to identify. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

Fig. 2 shows the data and results. As r increases the anomalies become more similar to the non-anomalies, so the true positive rate falls. The new version performs better than the original at every value of r , with the largest differences at the smallest values of r , where the anomalies are farthest from the non-anomalous observations.

4.2 Experiment 2: Anomalies on the boundary as n increases

Anomalies are often close to the non-anomalous observations rather than well separated from them. Here the non-anomalous observations have independent $\mathcal{N}(0, 1)$ coordinates in $\mathbb{R}^2$, with $n \in \{1000, 2000, \dots, 10000\}$, and there are $0.005n$ anomalies in a ring just beyond most of them. For each anomaly we sample $\mu_{i1} \sim \mathcal{U}(-2.2\sqrt{2}, 2.2\sqrt{2})$, set $\mu_{i2} = \pm\sqrt{2 \times 2.2^2 - \mu_{i1}^2}$ with the negative sign for a randomly chosen half of the anomalies, so that the anomaly means lie on a circle of radius $2.2\sqrt{2}$ about the origin, and draw $Y_{i1} \sim \mathcal{N}(\mu_{i1}, 0.1)$ and $Y_{i2} \sim \mathcal{N}(\mu_{i2}, 0.1)$. There are 10 replications for each n , and for both versions the kernel density estimates use the truncated sum described in Section 3, with 100 neighbours.

Fig. 3 shows the data of one replication and the results. Both versions have low false positive rates. The new version has a higher true positive rate at the smaller sample sizes, and the two are similar beyond about $n = 5000$. The online Appendix repeats this experiment with gamma-distributed observations (Experiment S2); there the two versions differ little, and the original is slightly better at the largest sample sizes.

4.3 Experiment 3: Normal distributions moving out in each iteration

The sample has 405 points in $\mathbb{R}^6$, of which five are anomalies. The non-anomalies are $\mathcal{N}(0, 1)$ in every dimension. The anomalies differ from the non-anomalies only in the first dimension, where they are $\mathcal{N}(2 + (i - 1) \times 0.5, 0.2)$ at iteration $i \in \{1, \dots, 10\}$, so they move further from the non-anomalous points at each iteration. Each iteration is repeated 10 times.

Fig. 4 shows the data and the results, with anomalies in red. The top left plot shows the $Y_1 Y_2$

plane at the third iteration, and the middle plots the Y_1Y_2 and Y_3Y_4 planes at the ninth, by which time the anomalies have separated from the other points in the Y_1 direction but remain mixed with them on the Y_3Y_4 plane. The right panel gives the AUC, Fmeasure and Gmean. The new version has higher Fmeasure and Gmean than the original from about the third iteration, the two versions have the same AUC, and both are at least as good as the four other methods on every metric.

4.4 Experiment 4: Annulus in $\mathbb{R}^3$

The sample has 805 points in $\mathbb{R}^3$, of which five are anomalies. The non-anomalies lie in an annulus on the Y_1Y_2 plane with $Y_3 \sim \mathcal{U}(0, 1)$, and the anomalies move towards the centre of the hole over

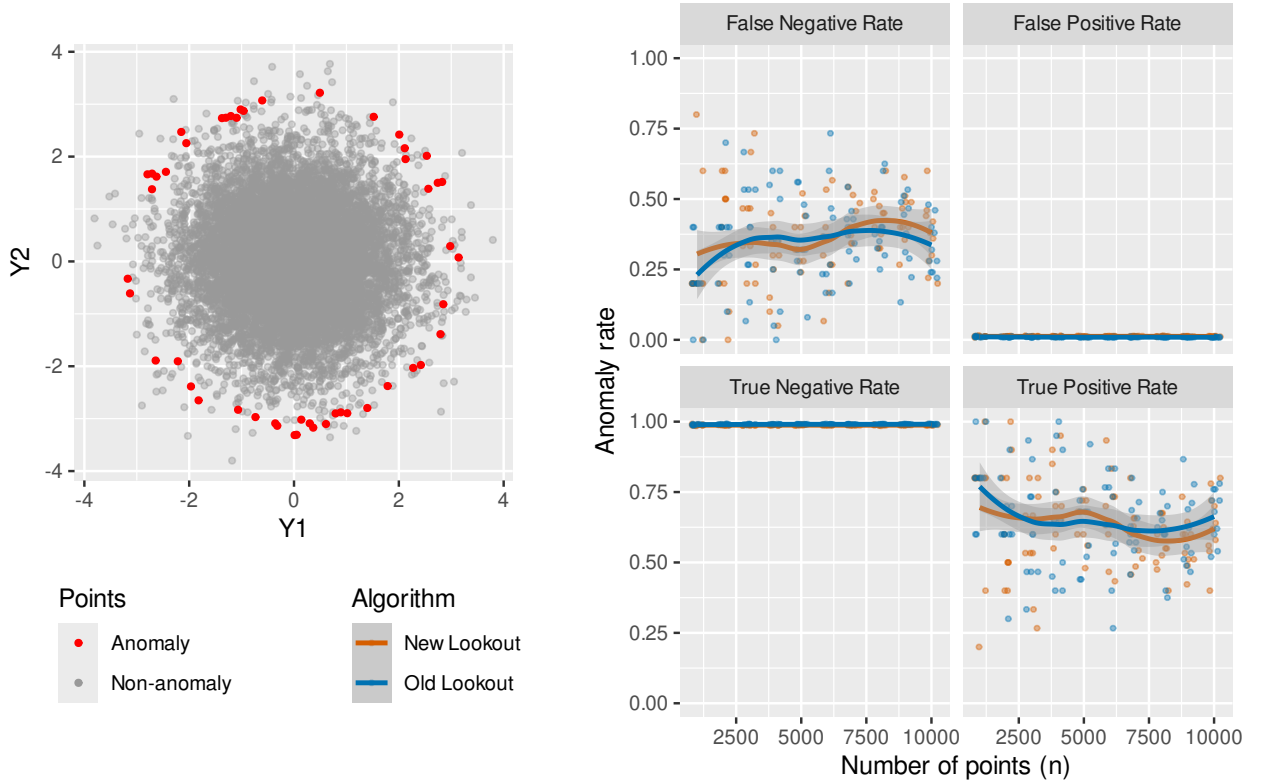


Figure 3: Experiment 2 data and results using Normal distributions. Left: data for $n = 10000$. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

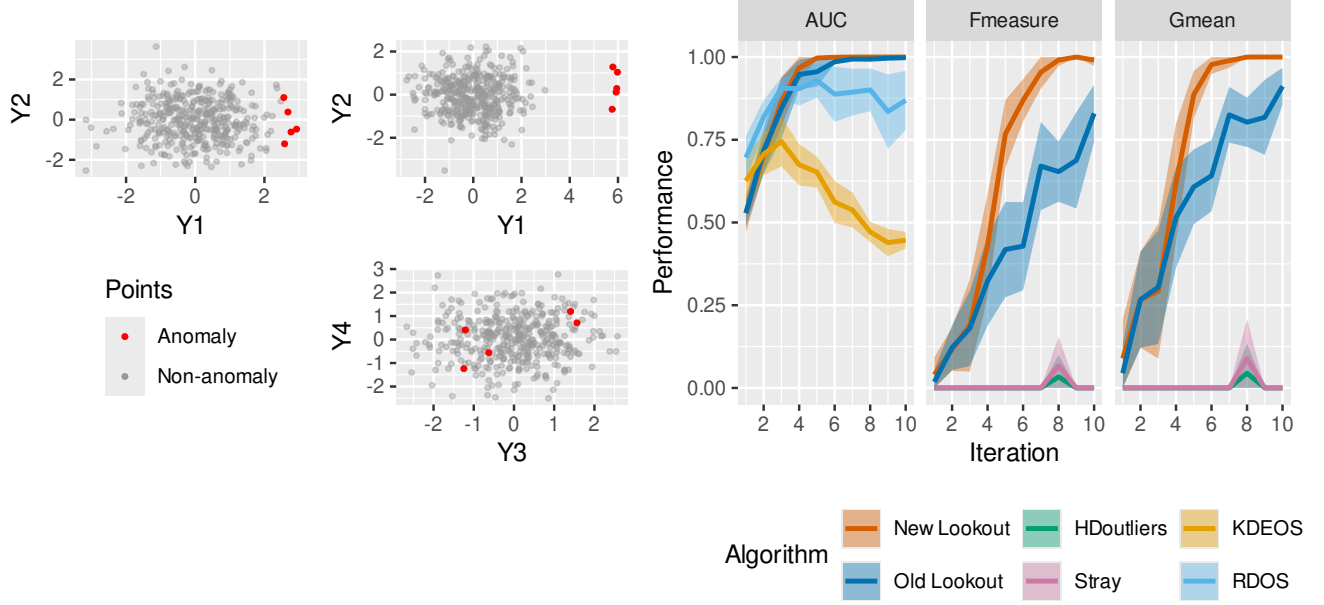


Figure 4: Experiment 3 results. Top left: Data from the 3rd iteration showing the Y_1Y_2 plane with anomalies in red. Middle left: Data from the 9th iteration showing both the Y_1Y_2 plane and Y_3Y_4 plane, with anomalies moving further away in the Y_1 direction, resulting in the anomalies and non-anomalies being mixed on the Y_3Y_4 plane. Right: AUC, Fmeasure and Gmean of old and new versions of lookout, HDoutliers, KDEOS, RDOS and stray.

10 iterations, each repeated 10 times. Fig. 5 shows the data at iterations 3 and 9 and the results. The new version identifies all five anomalies from the fifth iteration, with AUC, Fmeasure and Gmean equal to one. The original version is lower and more variable, KDEOS and RDOS have AUC of at most about 0.6 throughout, and HDoutliers and stray identify no anomalies until about the eighth iteration.

4.5 Experiment 5: Unit cube in $\mathbb{R}^{20}$

The sample has 500 points in $\mathbb{R}^{20}$, of which one is an anomaly. Initially all 500 points are uniform on $(0,1)$ in each dimension. At iteration $i \in \{1, \dots, 20\}$ the first i coordinates of the anomaly are set to 0.9, so that at iteration 20 it is $(0.9, \dots, 0.9)$, and each iteration is repeated 10 times.

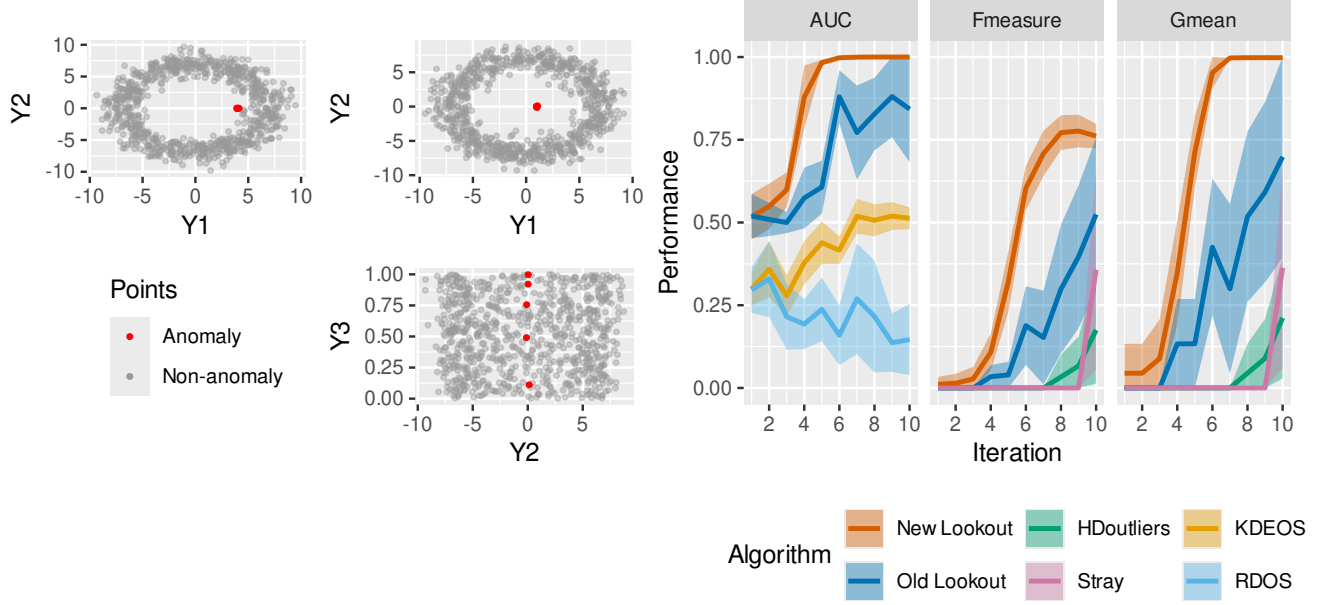


Figure 5: Experiment 4 results. Top left: Data from the 3rd iteration showing the Y_1Y_2 plane with anomalies in red. Middle left: Data from the 9th iteration showing both the Y_1Y_2 plane and Y_2Y_3 plane, with anomalies moving into the annulus. Right: AUC, Fmeasure and Gmean of old and new versions of lookout, HDoutliers, KDEOS, RDOS and stray.

The anomaly is not anomalous in any single dimension or on any Y_iY_j plane, nor in the space of the first two principal components. To display it we use *dobin* (Kandanaarachchi & Hyndman 2021), a dimension reduction method designed for anomalies. Fig. 6 shows the D_1D_2 plane of the *dobin* projection at iterations 5, 12 and 20, where the anomaly is clearly separated only at the last, together with the results. Both versions of lookout are run without scaling, and they perform identically. Their AUC reaches one by about the twelfth iteration, as does that of RDOS, with KDEOS a little lower. Gmean reaches one soon after; Fmeasure stays below 0.5, because a single false positive halves it when there is one anomaly. HDoutliers and stray never identify the anomaly.

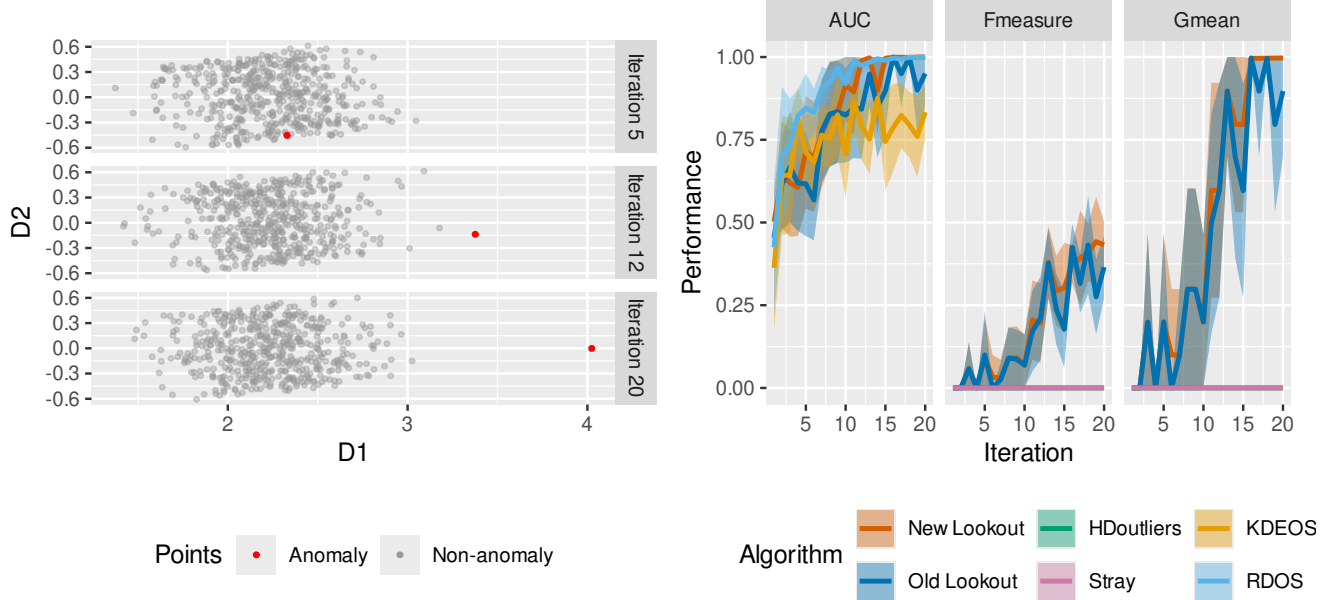


Figure 6: Experiment 5 results. Left: Data from the 5th, 12th and 20th iterations, showing the D_1D_2 plane from the dobin projection. Anomalies are shown in red. Right: AUC, Fmeasure and Gmean of old and new versions of lookout, HDoutliers, KDEOS, RDOS and stray.

4.6 Sensitivity to the bandwidth

Section 2.6 argues that the bandwidth should be of the order of the spacing between observations in the sparse part of the sample, and that neither the constant nor MISE-optimality matters much. To check this, we repeat Experiments 1, 2 and 3, and Experiment S1 of the online Appendix, with the bandwidth h_n multiplied by $1/4$, $1/2$, 2 and 4 , and with h_n replaced by the bandwidth h_{opt} of Table 2, which is the AMISE-optimal bandwidth for the Epanechnikov kernel and a standard normal density, computed for the standardized data. Everything else in Algorithm 2 is unchanged. In Experiment 2 we use $n \in \{1000, 2000, 4000\}$ and the exact leave-one-out estimate rather than the truncated sum described in Section 3, so that the results at large multiples of h_n are not affected by the truncation. Each setting is replicated ten times. Fig. 7 reports the area under the ROC curve computed from the anomaly scores, and the true and false positive rates at $\alpha = 0.01$.

The ranking of the observations is insensitive to the bandwidth. The area under the ROC

curve is almost the same for every bandwidth from $h_n/2$ to $4h_n$ in all four experiments, a range of a factor of eight, and falls at $h_n/4$ in Experiments 2, 3 and S1. The threshold is more sensitive. At h_n the false positive rate is between 0.005 and 0.013 in the four experiments, close to the nominal $\alpha = 0.01$. Halving the bandwidth roughly doubles it, and at $h_n/4$ it is between 5% and 27%, because observations with no other observation inside the kernel support have a leave-one-out estimate of zero and are all declared anomalous. Doubling or quadrupling the bandwidth keeps the false positive rate at or below the nominal level and, in Experiments 1 and 2, lowers the true positive rate, since the estimate at an isolated observation then includes contributions from the bulk of the sample. In Experiments 2 and S1, where the non-anomalous observations are normal in two dimensions, h_{opt} is close to h_n and about $0.75h_n$ respectively, and gives results similar to h_n . In Experiment 1, where the observations are gamma distributed, h_{opt} is about a third of h_n on average, and in Experiment 3, where $m = 6$, about a quarter. In Experiment 1 its false positive rate is 2%. In Experiment 3 the GPD could not be fitted in 73 of the 100 runs, because more than 10% of the observations had no other observation inside the kernel support, and in the remaining runs the false positive rate averaged 0.34. This is the behaviour described in [Section 2.6](#): in moderate dimensions a bandwidth chosen for the bulk of the distribution is too small for the tails.

5 Applications

We apply the new and original versions of lookout to two real datasets. Neither has labelled anomalies, so accuracy cannot be measured, but both have two variables, so the output can be inspected.

1. Old Faithful dataset¹: 2,097 eruptions of the Old Faithful Geyser in Yellowstone National Park, Wyoming, USA, from 14 January 2017 to 29 December 2023. For each recorded erup-

¹Data downloaded from <https://geysertimes.org>.

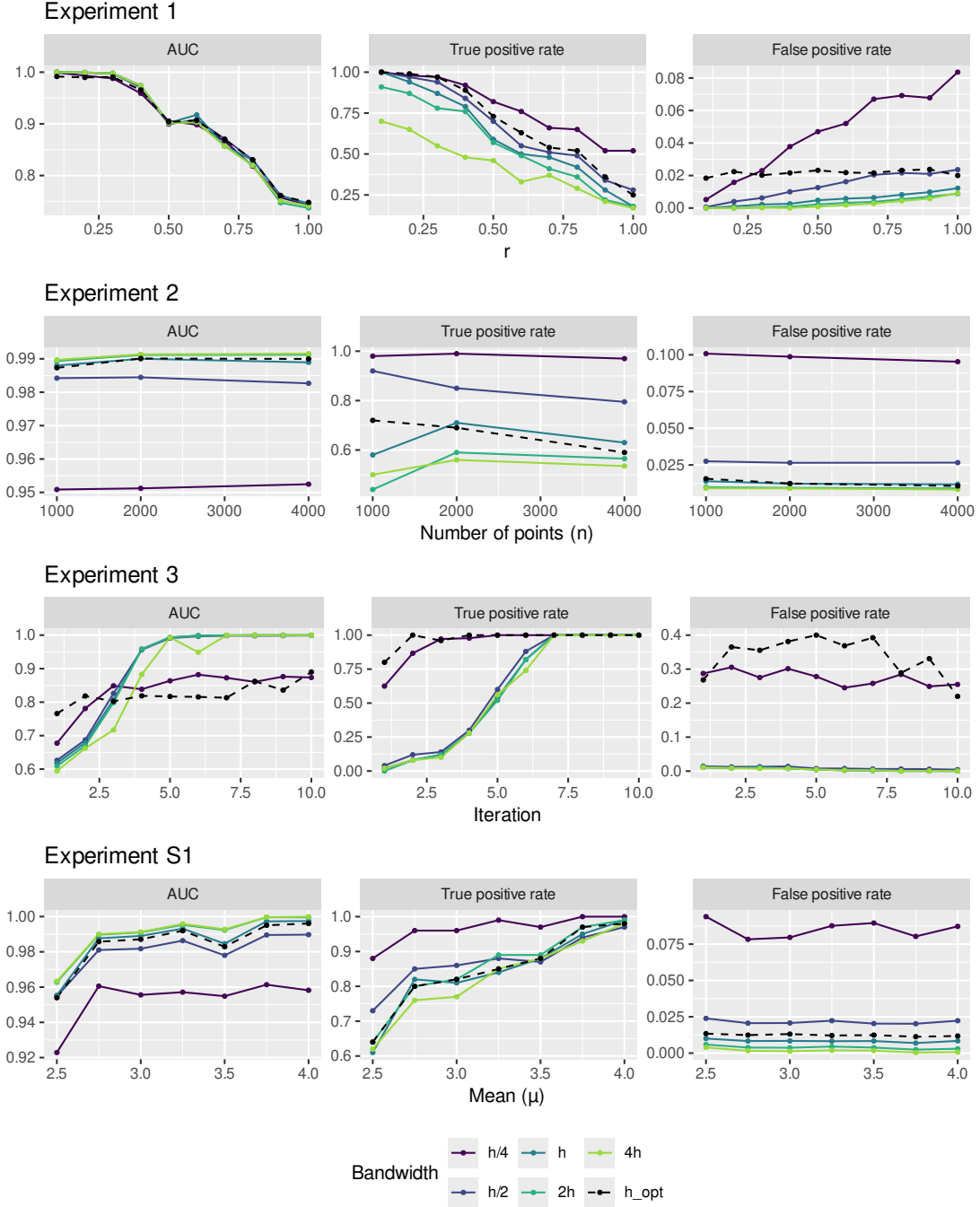


Figure 7: Sensitivity of the modified lookout algorithm to the bandwidth in Experiments 1, 2 and 3 and in Experiment S1 of the online Appendix. Each panel shows the mean over ten replicates of the area under the ROC curve, the true positive rate or the false positive rate, with the minimum spanning tree bandwidth h_n multiplied by 1/4, 1/2, 1, 2 and 4, and with the AMISE-optimal bandwidth h_{opt} of Table 2 (dashed).

tion the duration is given in seconds along with the waiting time to the following eruption. The records are incomplete, especially during winter.

2. Wine reviews dataset²: 110,203 reviews of wines from 42 countries, obtained from the *Wine Enthusiast Magazine* during the week of 15 June 2017. We use the 4,496 wines labelled Shiraz or Syrah, and two variables: the points score given by the *Wine Enthusiast* reviewer on a scale of 0–100, and the price of a bottle. The kernel density estimates use the truncated sum described in Section 3, with 100 neighbours.

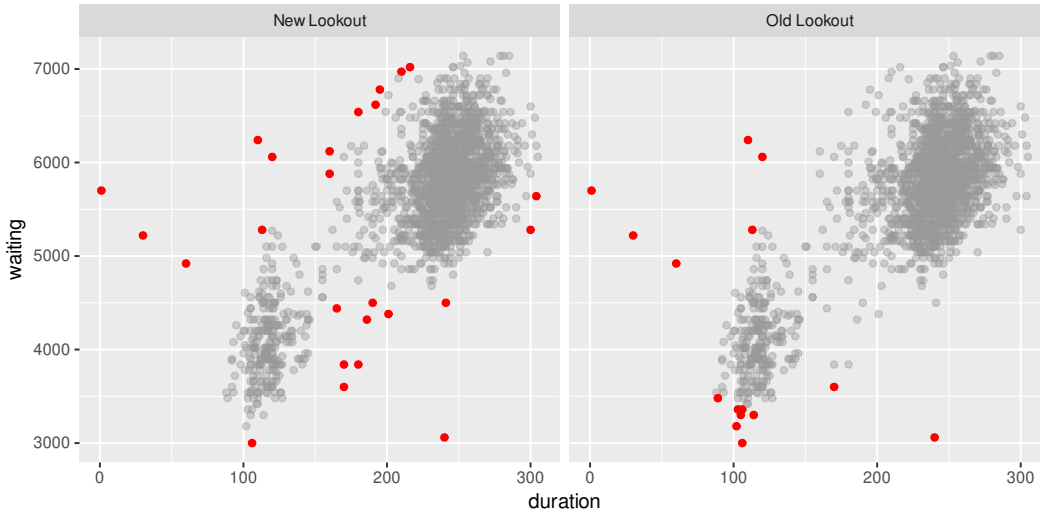


Figure 8: Anomalies (red) identified in the Old Faithful data by the new version of lookout (left) and the original version (right).

Fig. 8 shows the Old Faithful data. The new version identifies the observations in the sparse regions between and around the two clusters. The original identifies fewer of those, and instead flags a group of observations at the lower edge of the smaller cluster, close to the bulk of the data.

Fig. 9 shows the wine reviews. The new version identifies the expensive wines and little else. The original also flags many wines with the highest and lowest scores at ordinary prices, and several at the edge of the main cloud. In both datasets the new version flags the observations that

²Data downloaded from <https://www.kaggle.com/datasets/zynicide/wine-reviews/data>.

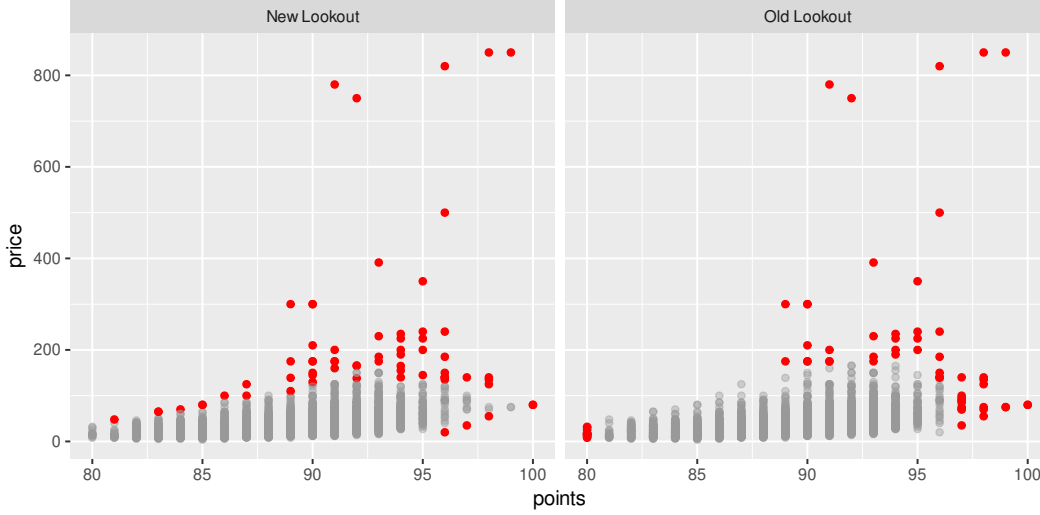


Figure 9: Anomalies (red) identified in the wine reviews data by the new version of lookout (left) and the original version (right).

are isolated in the two-dimensional space, and the original flags observations at the margins of the dense regions as well; we regard the former as the more useful behaviour.

6 Discussion and conclusions

The bandwidth of the lookout algorithm was defined through the death diameters of a Vietoris-Rips filtration, which are the edge lengths of the Euclidean minimum spanning tree of the data. Working with the tree has allowed us to analyse the bandwidth and to modify the algorithm. The modified algorithm takes an upper quantile of the edge lengths as its bandwidth in place of the largest gap between consecutive edge lengths. The scaled quantile converges in probability to a constant for any density, so the bandwidth has the order of the spacing between neighbouring observations, whereas the longest edges of the tree grow without bound under heavy tails and shrink only as $\log \log n / \sqrt{\log n}$ for Gaussian data. The resulting kernel density estimator is not MISE-consistent, and at practical sample sizes in moderate dimensions its bandwidth exceeds the MISE-optimal one, because the quantile is set by the spacing in the sparse part of the sample, where anomalies are identified. In [Section 4.6](#) the ranking of the observations was insensitive to the

bandwidth over a factor of eight, while the MISE-optimal bandwidth left too many observations isolated in six dimensions for the generalized Pareto distribution to be fitted. The two other modifications are a robust standardization with the minimum covariance determinant estimator, and a non-positive shape parameter in the generalized Pareto fit.

In the simulations the modified algorithm has a higher true positive rate than the original when the anomalies are close to the bulk of the data (Experiments 1 to 4), the same performance when they are well separated (Experiment 5, and Experiment S1 of the online Appendix), and a slightly lower true positive rate at the largest sample sizes of Experiment S2 of the online Appendix. It is at least as good as HDoutliers, stray, KDEOS and RDOS on every metric in Experiment 3, ahead of all four in Experiment 4, and in Experiment 5 it matches RDOS on AUC while HDoutliers and stray never identify the anomaly. On the Old Faithful and wine reviews data it identifies the observations in the sparse regions, and fewer observations at the edge of the main cloud than the original.

The algorithm applies to any continuous multivariate data, and the experiments show good results up to 20 dimensions. It can be applied to problems that can be transformed to that setting, such as image anomaly detection through feature extraction or functional data through functional principal components. It assumes independent and identically distributed observations, so temporal or spatial data need pre-processing, for example by applying the algorithm to the residuals of a fitted model. In high dimensions kernel density estimation suffers from the curse of dimensionality, and dimension reduction such as principal components or dobin ([Kandanaarachchi 2022](#)) before applying lookout may help.

The minimum spanning tree is computed by the dual-tree Borůvka algorithm of [March et al. \(2010\)](#) in $O(n \log n)$ time under mild conditions, and takes a few seconds for $n = 10^6$ in two dimensions and about a minute for $n = 10^5$ in ten ([Table 1](#)). The kernel density estimate dominates the cost. Its exact computation uses a fixed-radius search, at a cost proportional to the number of pairs of observations inside the kernel support. That number is small in two dimensions, but

in three or more the support in the bulk of the data holds a large fraction of the sample, so the exact estimate is practical only for n up to about 10^4 , and beyond that the implementation uses a truncated sum over nearest neighbours, which takes two and a half minutes for $n = 10^5$ in ten dimensions. Two theoretical questions remain open. We have no formula for where the largest gap between consecutive edge lengths occurs, so the bandwidth of the original algorithm has not been analysed beyond the results on the longest edge. And the constant $q_\gamma(f)$ of [Proposition 1](#) is defined through the component count of a Poisson process and is not explicit, although the bounds t_- and t_+ are.

Supplementary material

The online Appendix contains the proofs of [Proposition 1](#) and [Counterexample 1](#), two further comparisons of the original and modified algorithms, and a set of showcase examples. The updated algorithm is implemented in the `lookout` R package ([Kandanaarachchi et al. 2026](#)), available on CRAN at <https://CRAN.R-project.org/package=lookout>. Files to reproduce all numerical examples are available at <https://github.com/robjhyndman/lookout2>.

Funding

Rob Hyndman and Sevvandi Kandanaarachchi are part of the Australian Research Council Industrial Transformation Training Centre in Optimisation Technologies, Integrated Methodologies, and Applications (OPTIMA), Project ID IC200100009. Katharine Turner is supported by an Australian Research Council Future Fellowship, FT250100951.

Disclosure statement

The authors report there are no competing interests to declare.

References

- Adams, H., Aminian, M., Farnell, E., Kirby, M., Mirth, J., Neville, R., Peterson, C. & Shonkwiler, C. (2020), A fractal dimension for measures via persistent homology, *in* ‘Topological Data Analysis: The Abel Symposium 2018’, Vol. 15 of *Abel Symposia*, Springer, pp. 1–31.
- Adler, R. J., Bobrowski, O. & Weinberger, S. (2014), ‘Crackle: The homology of noise’, *Discrete & computational geometry* **52**, 680–704.
- URL:** <https://doi.org/10.1007/s00454-014-9621-6>
- Ahmed, I., Dagnino, A. & Ding, Y. (2019), ‘Unsupervised anomaly detection based on minimum spanning tree approximated distance measures and its application to hydropower turbines’, *IEEE Transactions on Automation Science and Engineering* **16**(2), 654–667.
- Barnett, V. & Lewis, T. (1978), *Outliers in Statistical data*, John Wiley & Sons, Chichester, UK.
- Bommier, E. (2014), Peaks-over-threshold modelling of environmental data, Master’s thesis, Department of Mathematics, Uppsala University.
- Breunig, M. M., Kriegel, H.-P., Ng, R. T. & Sander, J. (2000), ‘LOF: identifying density-based local outliers’, *ACM Sigmod Record* **29**(2), 93–104.
- Caroni, C. & Prescott, P. (1995), ‘On Rohlf’s method for the detection of outliers in multivariate data’, *Journal of Multivariate Analysis* **52**(2), 295–307.
- Chacón, J. E. & Duong, T. (2018), *Multivariate kernel smoothing and its applications*, CRC Press, Boca Raton, Florida.
- Chaudhuri, D., Chaudhuri, B. B. & Murthy, C. A. (1996), ‘A data driven procedure for density estimation with some applications’, *Pattern Recognition* **29**(10), 1719–1736.
- Coles, S. (2001), *An introduction to statistical modeling of extreme values*, Vol. 208 of *Springer Series in Statistics*, Springer, London UK.

- Friedman, J. H. & Rafsky, L. C. (1979), ‘Multivariate generalizations of the Wald-Wolfowitz and Smirnov two-sample tests’, *The Annals of Statistics* **7**(4), 697–717.
- Ghrist, R. (2008), ‘Barcodes: the persistent topology of data’, *Bulletin of the American Mathematical Society* **45**(1), 61–75.
- Good, I. J. (1952), ‘Rational decisions’, *Journal of the Royal Statistical Society. Series B, Statistical methodology* **14**, 107–114.
URL: <http://dx.doi.org/10.1111/j.2517-6161.1952.tb00104.x>
- Gower, J. C. & Ross, G. J. S. (1969), ‘Minimum spanning trees and single linkage cluster analysis’, *Journal of the Royal Statistical Society. Series C, Applied Statistics* **18**, 54–64.
- Hawkins, D. M. (1980), *Identification of outliers*, Chapman & Hall, London, UK.
- Hsing, T. & Rootzén, H. (2005), ‘Extremes on trees’, *The Annals of Probability* **33**(1), 413–444.
- Hyndman, R. J. & Fan, Y. (1996), ‘Sample quantiles in statistical packages’, *The American Statistician* **50**, 361–365.
- Hyndman, R. J. & Frazier, D. T. (2026), Anomaly detection using surprisals, Working paper, Department of Econometrics & Business Statistics, Monash University.
URL: <http://robjhyndman.com/publications/surprisals.html>
- Jiang, M. F., Tseng, S. S. & Su, C. M. (2001), ‘Two-phase clustering process for outliers detection’, *Pattern Recognition Letters* **22**(6-7), 691–700.
- Juszczak, P., Tax, D. M. J., Pekalska, E. & Duin, R. P. W. (2009), ‘Minimum spanning tree based one-class classifier’, *Neurocomputing* **72**(7–9), 1859–1869.
- Kandanaarachchi, S. (2022), *dobin: Dimension Reduction for Outlier Detection*. R package v1.0.4.
URL: <https://CRAN.R-project.org/package=dobin>

- Kandanaarachchi, S. & Hyndman, R. J. (2021), ‘Dimension reduction for outlier detection using DOBIN’, *Journal of Computational and Graphical Statistics* **30**(1), 204–219.
URL: <https://doi.org/10.1080/10618600.2020.1807353>
- Kandanaarachchi, S. & Hyndman, R. J. (2022), ‘Leave-one-out kernel density estimates for outlier detection’, *Journal of Computational and Graphical Statistics* **31**(2), 586–599.
- Kandanaarachchi, S., Hyndman, R. J. & Fraley, C. (2026), *lookout: Leave One Out Kernel Density Estimates for Outlier Detection*. R package v2.1.0.
URL: <https://sevvandi.github.io/lookout/>
- Kolupuri, S. V. J., Paul, A., Bhowmick, R. S. & Ganguli, I. (2025), Scams and frauds in the digital age: ML-based detection and prevention strategies, *in* ‘Proceedings of the 26th International Conference on Distributed Computing and Networking’, pp. 340–345.
- Kruskal, J. B. (1956), ‘On the shortest spanning subtree of a graph and the traveling salesman problem’, *Proceedings of the American Mathematical Society* **7**(1), 48–50.
- March, W. B., Ram, P. & Gray, A. G. (2010), Fast Euclidean minimum spanning tree: algorithm, analysis, and applications, *in* ‘Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining’, ACM, New York, pp. 603–612.
- Mitsche, D. & Penrose, M. D. (2021), ‘Limit theory of combinatorial optimization for random geometric graphs’, *The Annals of Applied Probability* **31**(6), 2721–2771.
- Penrose, M. D. (1997), ‘The longest edge of the random minimal spanning tree’, *The Annals of Applied Probability* **7**(2), 340–361.
- Penrose, M. D. (1998), ‘Extremes for the minimal spanning tree on normally distributed points’, *Advances in Applied Probability* **30**(3), 628–639.

- Penrose, M. D. (2003), *Random Geometric Graphs*, Vol. 5 of *Oxford Studies in Probability*, Oxford University Press, Oxford.
- Pickands, J. (1975), ‘Statistical inference using extreme order statistics’, *The Annals of Statistics* **3**(1), 119–131.
- Rohlf, F. J. (1975), ‘Generalization of the gap test for the detection of multivariate outliers’, *Biometrics* **31**(1), 93–101.
- Rousseeuw, P. J. & Driessen, K. V. (1999), ‘A fast algorithm for the minimum covariance determinant estimator’, *Technometrics* **41**(3), 212–223.
- Schubert, E., Zimek, A. & Kriegel, H. P. (2014), ‘Generalized outlier detection with flexible kernel density estimates’, *SIAM International Conference on Data Mining 2014* pp. 542–550.
- Singh Parihar, Y., Curtin, R., Eddelbuettel, D. & Balamuta, J. (2026), *mlpack: Rcpp Integration for the mlpack Library*. R package version 4.8.0-1.
URL: <https://www.mlpack.org/doc/user/bindings/r.html>
- Sreevani & Murthy, C. A. (2016), ‘On bandwidth selection using minimal spanning tree for kernel density estimation’, *Computational Statistics & Data Analysis* **102**, 67–84.
- Stone, J. V. (2022), *Information theory: a tutorial introduction*, Sebtel Press, Sheffield, England.
- Talagala, P. D., Hyndman, R. J. & Smith-Miles, K. (2021), ‘Anomaly detection in high-dimensional data’, *Journal of Computational and Graphical Statistics* **30**(2), 360–374.
- Tang, B. & He, H. (2017), ‘A local density-based approach for outlier detection’, *Neurocomputing* **241**, 171–180.
- Wilkinson, L. (2017), ‘Visualizing big data outliers through distributed aggregation’, *IEEE Transactions on Visualization and Computer Graphics* **24**(1), 256–266.

Zahn, C. T. (1971), ‘Graph-theoretical methods for detecting and describing gestalt clusters’, *IEEE Transactions on Computers* **C-20**(1), 68–86.